Steering a swarm that cannot prove anything

Simulation 2 (asynchronous activation), setup `task003-nosolution`

LLM-free simulator; report of 17 October 2026; runs of 16 October 2026

Same structure as the gold-standard report on `task003-symmetric`; mirrors the Simulation 1 report of the same date.

How the simulator works. The world, `task_003`, has three candidate allocations; A0 is the correct one. Each of 24 agents is given one true fact and reasons as an exact Bayesian: its *belief* is the exact probability of A0, A1 and A2 given the facts it currently remembers. There are no rounds. Each agent acts at random times, on average once per time unit: it reads q posts drawn at random from the 48 most recent posts on a shared *board* (never its own), adds their facts to its memory, recomputes its belief, *votes* by probability matching (a vote drawn from its belief) and posts one remembered fact that supports its vote. Every remembered fact is forgotten after a random lifetime (on average a fraction ρ survives one time unit; reading it again renews it). A *controller* with a target, A0 (truth) or A2 (false), acts at random times at rate λ_c until $t = 30$: it reads q_c recent agent posts and, unless the *vote gate* (at least 75% of them from agents voting for its target) silences it, posts the one fact from one shared pool that best supports its target, at most 30 messages per episode. Agents continue alone until $t = 40$. Every episode is run three times with the same random draws: without a controller (*silent*), with the controller targeting A0 and with the one targeting A2.

The facts at the start. All facts are true, and no set of them proves any allocation: the agents' evidence is a coin flip between A0 and A2, and one pool serves both targets (Table 1).

Table 1: Facts available at $t = 0$ and what each set implies, $P(\text{A0}, \text{A1}, \text{A2} \mid \text{facts})$.

set of facts	facts	$P(\text{A0}, \text{A1}, \text{A2})$
agents' facts, all together (24 agents, one fact each)	15 distinct	(0.5, 0, 0.5): no proof
shared pool (both targets)	12, all new to the agents	(0.37, 0.26, 0.37)
agents' facts + pool	27 distinct	(0.75, 0, 0.25)

The runs. Posts read by an agent $q \in \{3, 6, 12\}$; posts read by the controller $q_c \in \{3, 6, 12, 24\}$; controller rate $\lambda_c \in \{0.5, 1, 2, 3, 4, 5, 6, 7, 8\}$ (one fact per message); memory $\rho \in \{1, 0.875, \ldots, 0.125\}$ (8 values; this report shows $\rho = 1, 0.75, 0.5, 0.25$ like the gold standard, all 8 are in the notebook `shared_analysis/gold_standard/explorer.ipynb`); vote gate on and off; 1,000 episodes per setting (3,480 settings). Unless stated otherwise: $q = 6$, $q_c = 12$, $\lambda_c = 1$, gate on. Cells shared with the 9 October grid reproduce it exactly.

Summary. The 24 agents' evidence is a coin flip between A0 (the truth) and A2, and no set of facts can prove either. **Without a proof to defend, every swarm is steerable**: either controller tips it within about ten time units at every memory, full memory included (Figure 1). The truth controller's susceptibility is 0.31–0.41 almost regardless of memory and rate; the false controller's grows as memory shortens (0.05–0.31), and **with full memory a faster false controller steers less** (Figure 2). The rate buys speed early and nothing by $t = 30$; per fact, slow control is the most effective (Figures 3, 7, 8). As in the symmetric setup, susceptibility and readability move together, the efficiency η stays at 0.82–0.93 except where the controllers become unequal, and reading more posts helps the controller when agents forget and hurts it with full memory (Figures 4–6).

1 Quantities

Votes. With $n_{a,t}$ the number of agents whose last vote is answer a at time t and N_t the number of agents who have voted,

$$x_{a,t} = n_{a,t}/N_t. \tag{1}$$

At $t = 0$ nobody has voted and we set $x_{a,0} = 0$; all agents have voted by $t \approx 5$. (The 9–14 October reports divided by 24, which counted agents who had not yet acted as voting against the target; this inflated the apparent rise of the shares and the apparent bias of the controller's sensing at early times.)

Paired susceptibility. Following `BLACKBOARD_OBSERVABLES_AND_EFFICIENCIES` (§3), for a controller policy π and an answer a,

$$\chi^{(a)}_{\pi,t} \equiv G^{(a)}_t(\pi) = \mathbb{E}\big[x^{\pi}_{a,t} - x^{\text{base}}_{a,t}\big], \tag{2}$$

the mean over episodes of the difference between the controlled run and the silent run of the same episode. We report it on the controller's own target: $G_t \equiv G_t^{(0)}(\pi^+)$ for the controller targeting A0 and $G_t \equiv G_t^{(2)}(\pi^-)$ for the one targeting A2.

Target and state. The target $Z \in \{\mathrm{A0}, \mathrm{A2}\}$ is drawn by a fair coin. The state X at time t is the vote of a randomly chosen agent, $X \in \{\mathrm{A0}, \mathrm{A1}, \mathrm{A2}\}$. Each episode e of arm s (silent 0, target A0 $+$, target A2 $-$) contributes its vote shares $y_s^{(e)}(x) = x_{x,t}$, and

$$P_s(x) = \frac{1}{E}\sum_{e=1}^{E} y_s^{(e)}(x), \qquad E = 1000. \tag{3}$$

Readability and efficiency. With $Q = \frac{1}{2}(P_+ + P_-)$ and $\mathrm{KL}(p\|q) = \sum_x p(x)\log_2[p(x)/q(x)]$,

$$I(Z;X) = \tfrac{1}{2}\,\mathrm{KL}(P_+\|Q) + \tfrac{1}{2}\,\mathrm{KL}(P_-\|Q), \tag{4}$$

the information the state carries about the target (at most 1 bit). The controllers' total change from silence splits exactly into a directed and an undirected part,

$$C = \tfrac{1}{2}\,\mathrm{KL}(P_+\|P_0) + \tfrac{1}{2}\,\mathrm{KL}(P_-\|P_0) = I(Z;X) + \mathrm{KL}(Q\|P_0), \tag{5}$$

and the efficiency is the directed share,

$$\eta = I(Z;X)/C. \tag{6}$$

Silence sometimes never shows a value of X that a controller reaches, which would make C infinite. Before computing any KL, every arm's distribution therefore receives the same pseudo-count of $\alpha = 0.5$ episodes per value of X (K values),

$$\tilde{P}_s(x) = \frac{E\,P_s(x) + \alpha}{E + \alpha K}. \tag{7}$$

With $\alpha = 1$ instead, η changes by at most 0.004; without it, by at most 0.005 where C is finite. η is not defined when $C < 0.01$ bits (no measurable change); such values are left blank. Equation (5) holds in the data to 4×10^{-15} bits.

Cost. F_t is the number of facts the controller has posted up to t (mean per episode); the gain per fact is G_t/F_t, and the marginal susceptibility is the slope between neighbouring rates,

$$\frac{\Delta G}{\Delta \lambda_c} = \frac{G(\lambda_c^{(k+1)}) - G(\lambda_c^{(k)})}{\lambda_c^{(k+1)} - \lambda_c^{(k)}}. \tag{8}$$

Episode e uses the same random draws at every rate, so the slope is estimated from per-episode differences, $[x^{(e)}_{\mathrm{target},t}(\lambda_c^{(k+1)}) - x^{(e)}_{\mathrm{target},t}(\lambda_c^{(k)})]/(\lambda_c^{(k+1)} - \lambda_c^{(k)})$, whose mean is Eq. (8) and whose standard error gives the 95% intervals (typically ± 0.01–0.03, about 30% narrower than for two independent means).

2 Votes over time

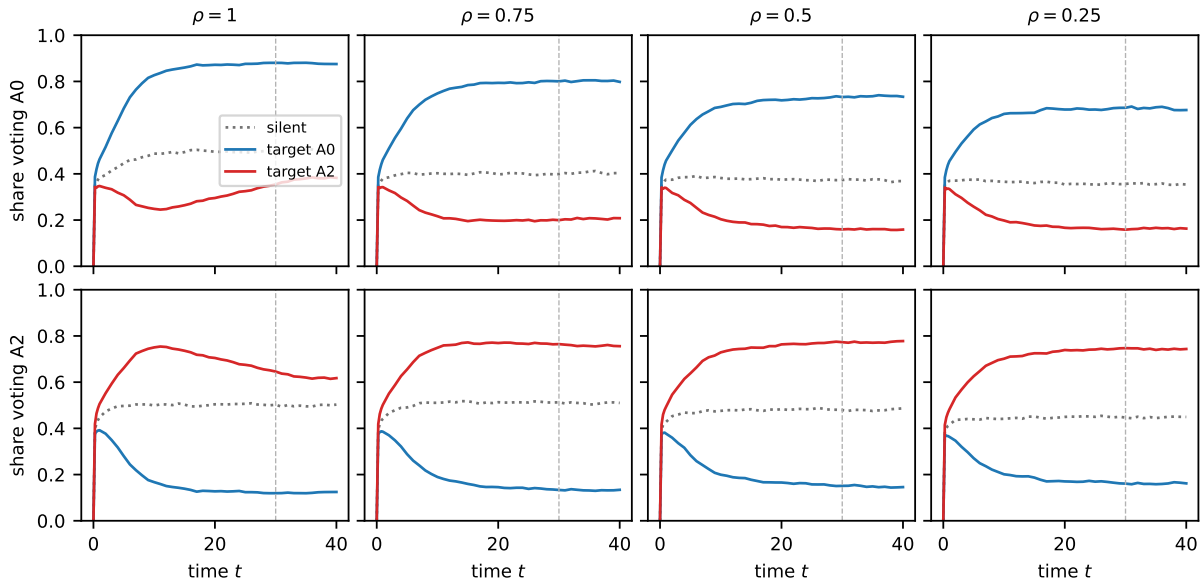


Figure 1: Share of agents voting A0 (top) and A2 (bottom), Eq. (1), without a controller (dotted), with the controller targeting A0 (blue) and A2 (red); one column per memory ρ. $q = 6$, $q_c = 12$, $\lambda_c = 1$, gate on. At $t = 0$ nobody has voted and every share is 0; after that the shares count only agents who have voted. The controller stops at $t = 30$ (dashed).

- **Without a controller the swarm stays split at every memory**: at $t = 30$, 36–50% vote A0 and 45–51% A2; when agents forget, it leans slightly to A2.
- **Either controller tips it within about ten time units, at every ρ**: the truth controller raises A0 to 0.88, 0.80, 0.73, 0.69 and the false controller A2 to 0.65, 0.76, 0.77, 0.75 for $\rho = 1, 0.75, 0.5, 0.25$.
- **Unlike in `task003-symmetric`, full memory does not protect the swarm**: there is no proof to rebuild, so control works without forgetting. Only the false controller's lead erodes with full memory (A2 at 0.75 at $t \approx 11$, 0.62 by $t = 40$), as agents accumulate evidence that favours A0.

3 Observables as time series

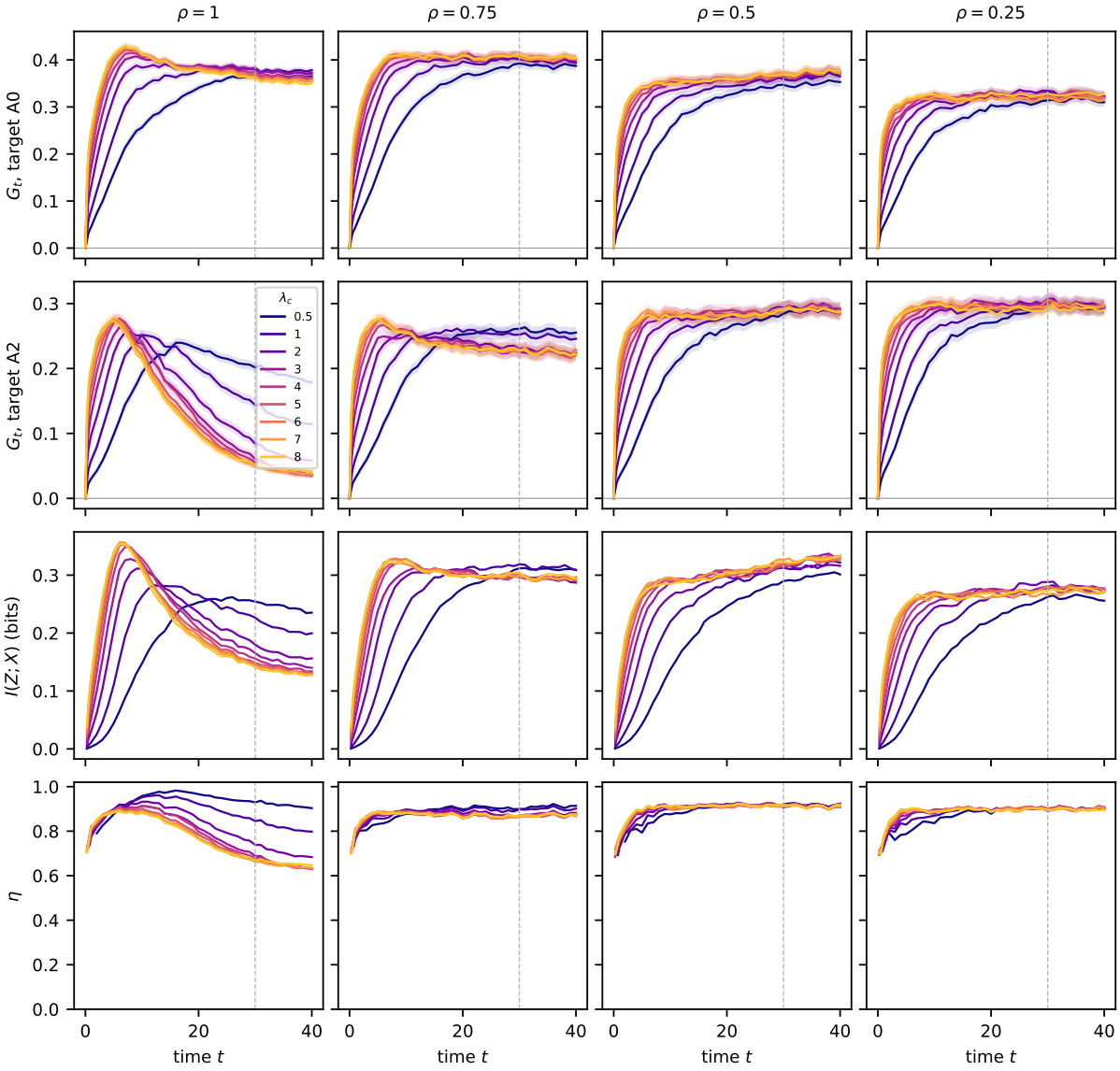


Figure 2: All observables over time, one column per memory ρ, one curve per controller rate λ_c; $q = 6$, $q_c = 12$, gate on. Rows: paired susceptibility G_t, Eq. (2), for target A0 and target A2 (bands: 95% intervals; $G_0 = 0$); readability $I(Z; X)$, Eq. (4); efficiency η, Eq. (6) (blank: undefined). The controller stops at $t = 30$ (dashed).

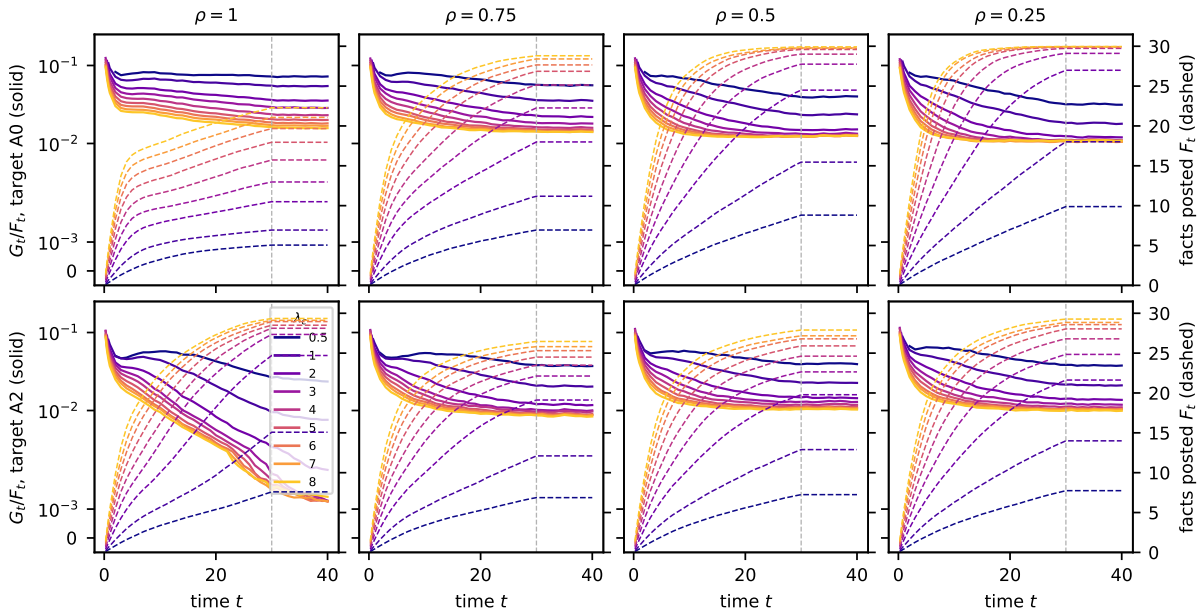


Figure 3: Gain per fact G_t/F_t (solid, left axis, log scale; shown once at least half a fact has been posted) and facts posted F_t (dashed, right axis), target A0 (top) and A2 (bottom), one colour per rate; $q = 6$, $q_c = 12$, gate on.

Susceptibility (Figure 2, rows 1–2).

- **The truth controller's susceptibility hardly depends on memory or rate**: at $t = 30$, $G = 0.36$–0.38, 0.39–0.41, 0.35–0.37, 0.31–0.33 for $\rho = 1, 0.75, 0.5, 0.25$.
- **The false controller's grows as memory shortens**: 0.05–0.20 ($\rho = 1$), 0.23–0.26, 0.28–0.29, 0.29–0.31.
- **With full memory, a faster false controller steers less**: its gain peaks early (0.27 at $t \approx 5$ for $\lambda_c = 8$) and falls to 0.05, while the slowest keeps 0.20. This is the rate reversal of the 9 October report. Its proposed explanation, not tested directly, is that posting more different pool facts pulls every belief toward the point all facts together imply, (0.75, 0, 0.25).
- **It persists after the controller stops**: from $t = 30$ to 40, G changes by at most 0.03.

Readability and efficiency (rows 3–4).

- **Readability is about the same at every** $\rho < 1$ ($I = 0.26$–0.33 bits at $t = 30$) and lower with full memory (0.14–0.25), where it falls with the rate as the false controller stalls.
- **Efficiency is flat when agents forget** ($\eta = 0.86$–0.92) and **falls with the rate with full memory** ($0.93 \to 0.66$): the pair turns lopsided, the truth controller keeps its gain and the false one loses it.

Gain per fact and facts posted (Figure 3).

- **The slowest controller is the most effective per fact**: at $t = 30$, target A2, $\rho = 0.5$: 0.039 per fact at $\lambda_c = 0.5$ (7 facts) against 0.010 at $\lambda_c = 8$ (28 facts); with full memory 0.027 against 0.002.
- **Both controllers spend most of their budget**: the swarm stays mixed, so the gate rarely silences either (target A0 posts 5–30 facts by $t = 30$, target A2 7–29).

4 Phase diagrams

All panels at $t = 30$, gate on; the susceptibility is for target A2 (grey number: its 95% interval includes 0); readability and efficiency are for the pair of targets.

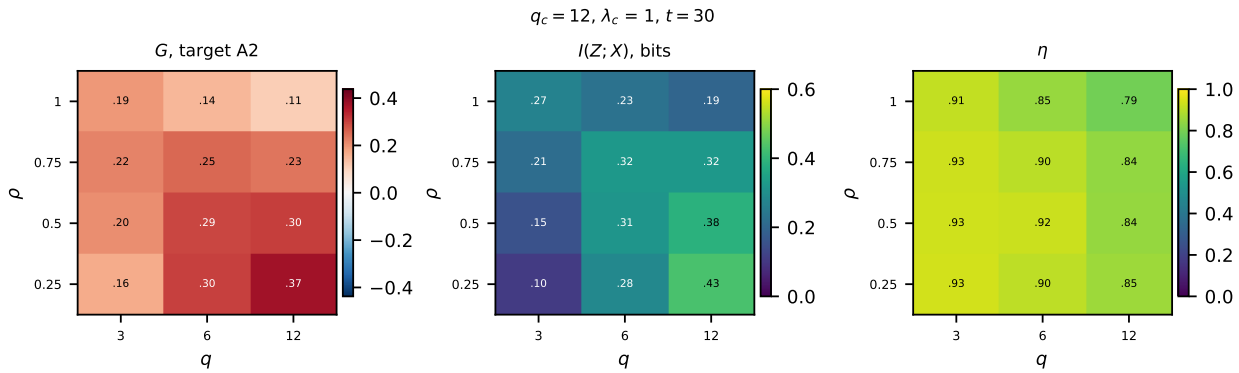


Figure 4: Susceptibility, readability and efficiency over memory ρ and the posts each agent reads q; $q_c = 12$, $\lambda_c = 1$.

- **Reading more helps the controller except with full memory**: at $\rho = 0.25$, G rises from 0.16 ($q = 3$) to 0.37 ($q = 12$) and I from 0.11 to 0.43 bits; at $\rho = 1$, G falls from 0.19 to 0.11 and I from 0.27 to 0.20. The same reversal as in `task003-symmetric`, at a different memory.
- η **stays at 0.79–0.93**, slightly lower at $q = 12$.

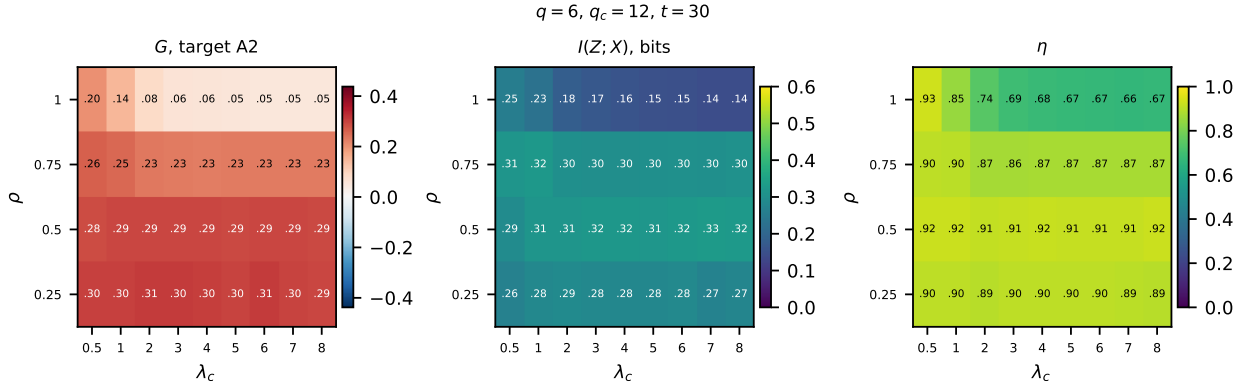


Figure 5: Susceptibility, readability and efficiency over memory ρ and the controller rate λ_c; $q = 6$, $q_c = 12$.

- **When agents forget, the rate changes nothing**: $G = 0.28$–0.31 ($\rho \leq 0.5$) and $I = 0.26$–0.33 bits along every row.
- **With full memory, faster is worse**: G falls from 0.20 ($\lambda_c = 0.5$) to 0.05 ($\lambda_c \geq 5$), I from 0.25 to 0.14 bits and η from 0.93 to 0.66.

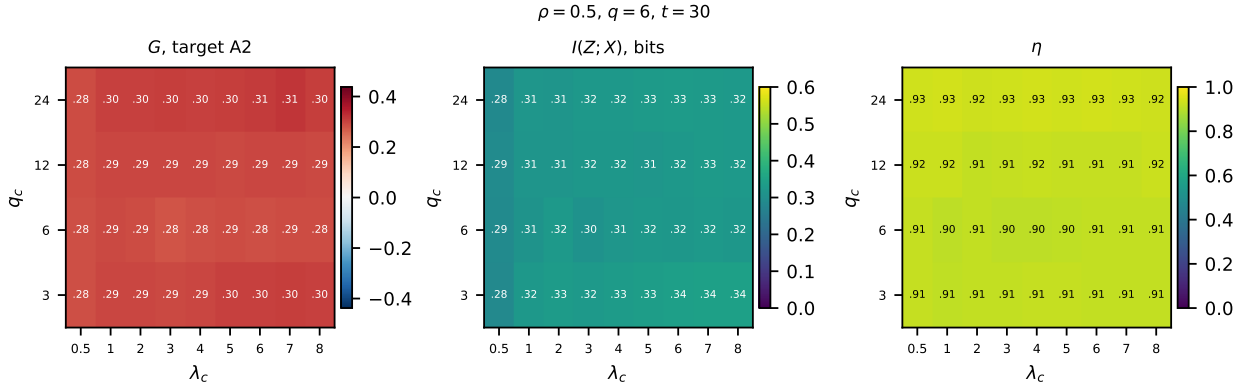


Figure 6: Susceptibility, readability and efficiency over the posts the controller reads q_c and its rate λ_c; $\rho = 0.5$, $q = 6$.

- **At $\rho = 0.5$ neither the controller's reading nor its rate matters**: $G = 0.28$–0.31 (mean 0.30, 0.28, 0.29, 0.30 for $q_c = 3, 6, 12, 24$), $I = 0.28$–0.34 bits, $\eta = 0.90$–0.93.

5 Susceptibility against the rate, and its slope

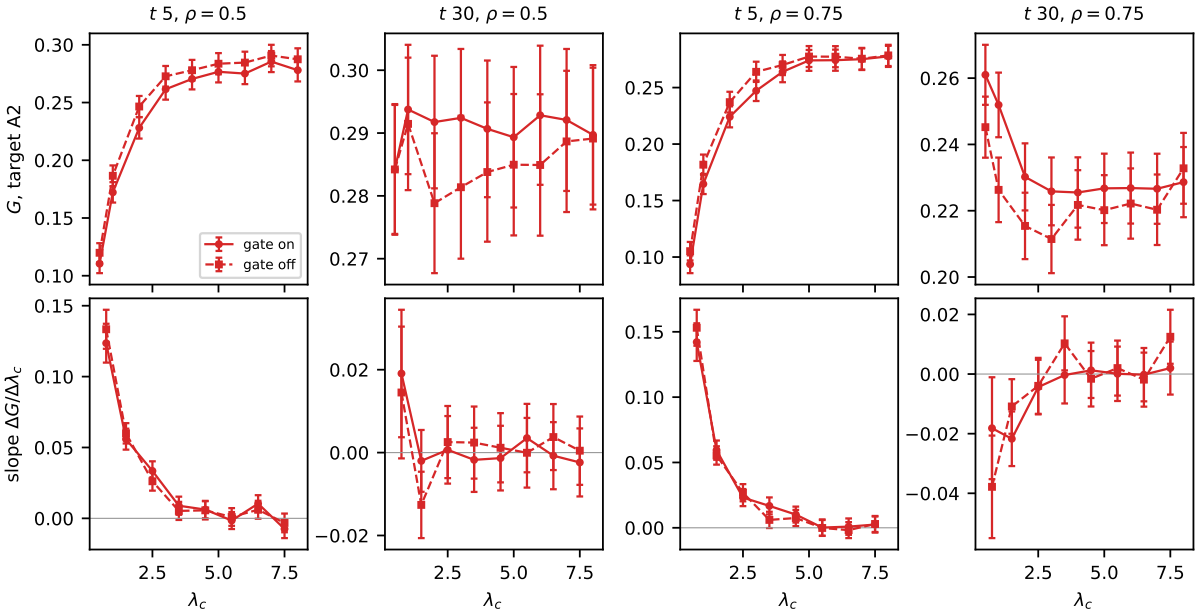


Figure 7: Top: G, target A2, against the controller rate λ_c, at $t = 5$ and $t = 30$ for $\rho = 0.5$ and 0.75; gate on (solid) and off (dashed); bars: 95% intervals. Bottom: the slope between neighbouring rates, Eq. (8), with paired 95% intervals.

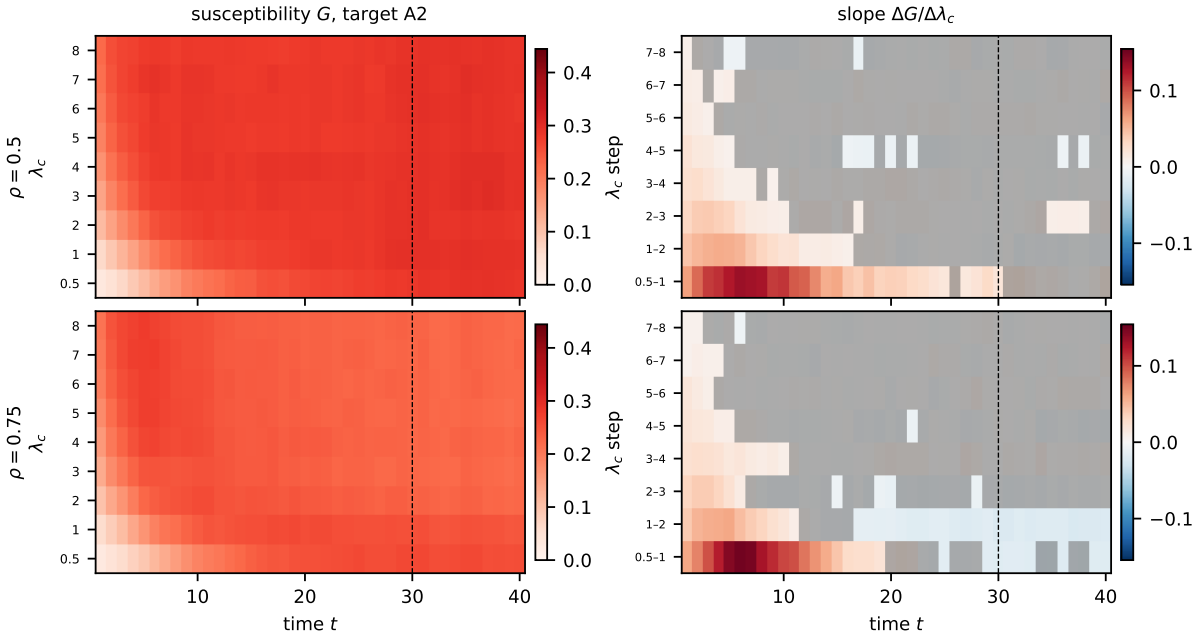


Figure 8: G, target A2 (left), and its slope between neighbouring rates (right) over time (x) and rate (y), for $\rho = 0.5$ (top) and 0.75 (bottom); $q = 6$, $q_c = 12$, gate on. Each row of the right panels is one step between neighbouring rates; faded cells: its paired 95% interval includes 0. The controller stops at $t = 30$ (dashed).

- **Early, speed buys susceptibility**: at $t = 5$ every step up to $\lambda_c \approx 5$ is significantly positive (at $\rho = 0.5$: 0.12 for $0.5 \to 1$, 0.06, 0.03, then about 0.01).
- **By $t = 30$ the advantage is gone**: at $\rho = 0.5$ only the first step is still positive (0.02); at $\rho = 0.75$ the first two are negative (-0.02 each), and with full memory they are clearly negative (-0.12, -0.06, -0.02 for the first three steps).
- **The slope map shows the crossover**: positive slopes at small rates and early times give way to faded (zero) or negative slopes after $t \approx 10$–15 (Figure 8).

- **Turning the gate off changes G by at most** ± 0.03 when agents forget, and lowers it by up to 0.08 with full memory.

No proofs here. In `task003-nosolution` no set of facts proves any allocation, so the gold standard's "who holds the proof" section has no counterpart.

6 Conclusions

1. **Without a proof to defend, every swarm is steerable.** Either controller tips it at every memory, including full memory; the truth controller gains 0.31–0.41, the false one 0.05–0.31.
2. **Memory now works against the false controller only with full memory**: the agents' own evidence then slowly pulls back toward A0, and a faster false controller does worse.
3. **The rate buys speed, not a larger end point**, and per fact slow control is the most effective.
4. **Susceptibility and readability move together; efficiency moves only where the two controllers become unequal** (full memory, fast rates).
5. **Reading more posts helps the controller when agents forget and hurts it with full memory.**
6. **Compared with Simulation 1** (same settings, about the same facts posted): the same picture, but Simulation 1's large budgets backfire, which a 30-message budget prevents here.

Caveats. One world, 24 exact-Bayesian agents voting by probability matching, one controller policy with a 30-message budget; 1,000 episodes per setting (95% intervals on G about ± 0.02). Information is plug-in from 1,000 episodes per arm. Data: `results/simulation_2/2026-10-16_sim2_nosolution_fine`; tables: `results/gold_standard/2026-10-16/`; code: `analysis/task003_llm_free/shared_analysis/gold_standard/`.